



iNSight Payment Gateway

**Bank to Agent's Wallet using
prefunded account**

How to interact with the iNSight payment gateway for
bank Interoperability

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1 INTRODUCTION

The *iNSight* Payment Gateway sits in front of a system that manages a customer's wallet. This wallet, in the case of mobile network operators, the "Mobile Money Platform" is physically separated from the payment gateway (sometimes also referred to as a "Broker" or an "Orchestration Layer") providing the benefit of a simplified interface and abstraction of both function and security.

11. BASIC FLOW

1.1.1 Prerequisites

In order to issue any query against the *iNSight* Payment Gateway, a valid session token is required.

The following data must be supplied by the owner of the system in order to be able to communicate with *iNSight* Payment Gateway.

- Service Provider Code
- URL for the webservice
- Username
- Password

The operation will be either Synchronous or Asynchronous depending on what is exposed by the Mobile Money platform (a platform that employs a Synchronous approach could be exposed via *iNSight* Payment Gateway as either Synchronous or Asynchronous. An Asynchronous platform can only be exposed in an Asynchronous approach).

12. CONNECTION DETAILS

1.1.2 Login

EventID	10033
Public IP	10.90.104.50
Internal IP	10.90.104.50
HTTP URL	http://10.90.104.50:8107/insight/B2BLogin

HTTPS URL	http://10.90.104.50:8107/insight/B2BLogin
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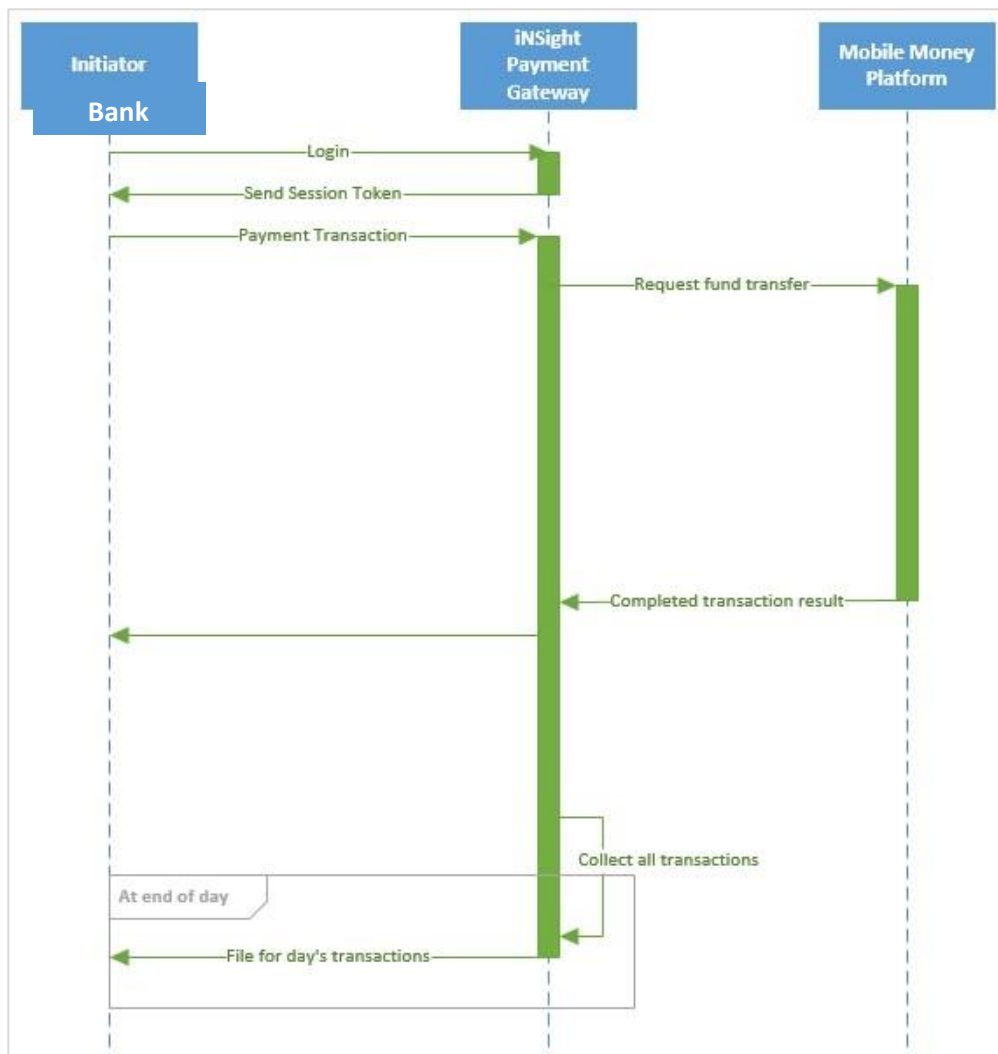
1.1.3 Event (e-money provisioning instruction)

EventID	10032
Public IP	10.90.104.50
Internal IP	10.90.104.50
HTTP URL	http://10.90.104.50:8106/insight/B2B
HTTPS URL	https://10.90.104.50:8110/insight/B2B

2 COMMUNICATION

2.1.1 Synchronous

The Synchronous approach is a much simpler one. Here, a request is sent by the initiator, with the response carrying the final status. In other words, once a response is received, the transaction has completed (with the response reporting the final status).

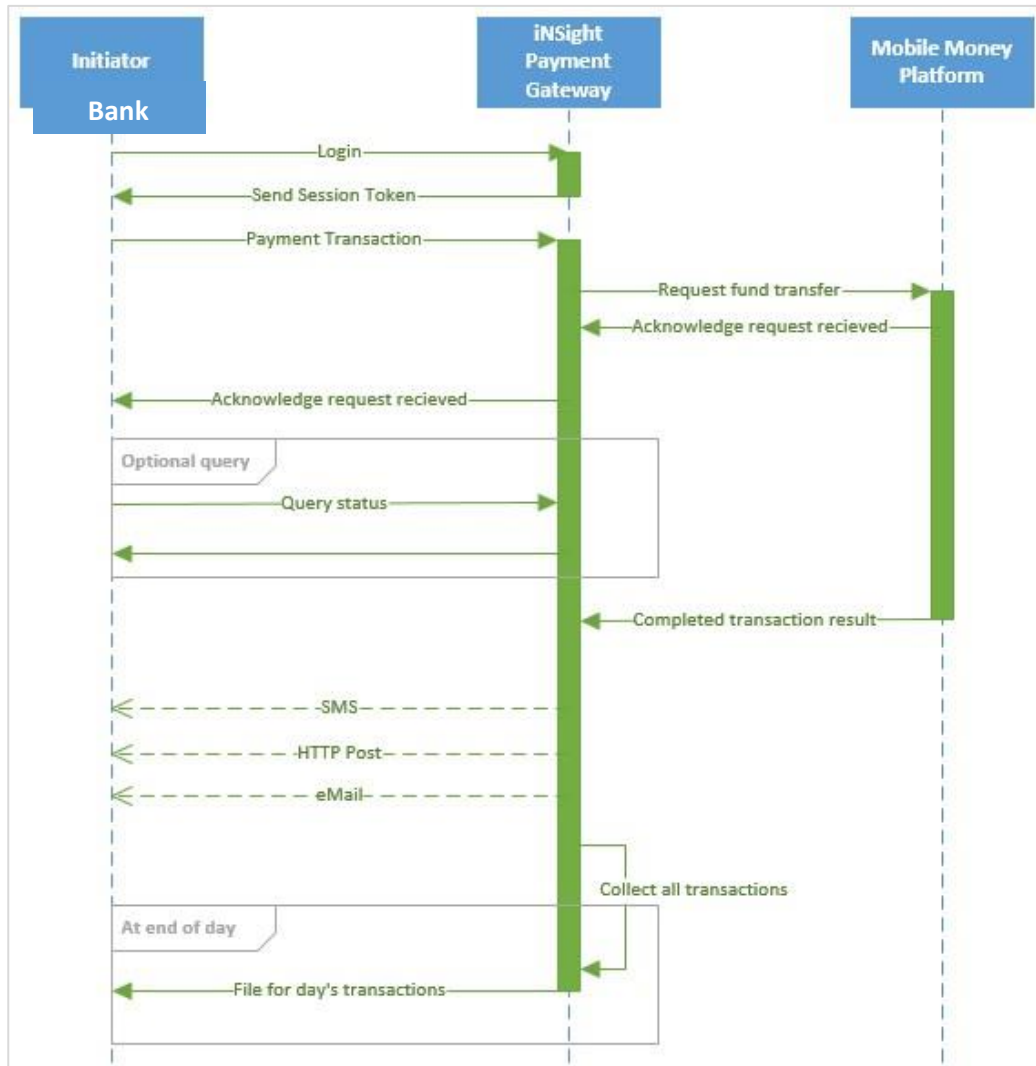


2.1.2 Asynchronous

With Asynchronous communication. While the request looks the same as with the Synchronous approach, the response that comes from the **iNSight** Payment Gateway is NOT the final status of the transaction request, it is simply an acknowledgement of receipt. The final status can be returned to the initiator via a number of channels:

- HTTP Post
 - o This requires that the initiator configure a webservice, accessible by the **iNSight** Payment Gateway. The webservice must conform to the specifications provided further down in this document.
- SMS
- Email

In addition to the transaction response, a file or report is composed at a configured interval (usually daily) and delivered to a destination agreed upon between the initiator and the **iNSight** Payment Gateway system owner.



3 LOGIN CALL (AUTHENTICATION)

Event Name	B2C Bank Interoperability
Authentication Required	Yes

This API call is used to receive a valid Session token to be passed when performing requests to iNSight Payment Gateway to authenticate the requestor.

13. REQUEST PARAMETERS

Name	Notes	Type
Username	Username of third party, provided by iPG	String Required

Password	Password of third party, provided by iPG	String Required
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14. RESPONSE PARAMETERS

Name	Notes	Type	Sample Values
SessionID	SessionID token to be used to authenticate requests.	String	cfb1b3b65d76438e83c6cd4fa1a17ab1
ResultCode	Code indicating the outcome of the API call.	String	0, -1
ResultDesc	Description for the corresponding code received	String	Success, Invalid Credentials

15. SAMPLE ENVELOPES

3.1.1 Login Request

```
<Request>
  <Username>user123</Username>
  <Password>test+123pas!</Password>
</Request>
```

3.1.2 Response

3.3.2.1 Success

```
<Response>
  <SessionID>cfb1b3b65d76438e83c6cd4fa1a17ab1</SessionID>
  <ResultCode>0</ResultCode>
  <ResultDesc>Success</ResultDesc>
</Response>
```

3.3.2.2 Invalid Credentials

```
<Response>
  <ResultCode>-1</ResultCode>
  <ResultDesc>Invalid Credentials</ResultDesc>
</Response>
```

4 B2B PAYMENT (BANK TO Agent's WALLET)

Event Name	B2B Bank To Wallet
Authentication Required	Yes

This event is called to request a transfer of funds from the Bank account (this account has to be prefunded) to the M-Pesa Business client /Short code M-Pesa account. The Bank uses this API call to move funds out of their M-Pesa account. This API call is **asynchronous**.

4.1 REQUEST PARAMETERS

Name	Notes	Type
Command	API call Identifier: BankB2W	String Required
SessionID	SessionID token used to authenticate requests	String Required
Shortcode	Business shortcode for wallet transfer to	String Required
BankShortcode	The unique bank identifier shortcode	String Required
Currency	Currency Code for the transaction. For example: CDF or USD	String Required
Amount	Amount of the said currency to transfer from the Bank to the customer	String Required
DateTime	Date of the transaction. Format YYYYMMDD HH24MiSS.	Date Optional
ThirdPartyReference	Bank's unique transaction reference number	String Required
TransactionID	Unique Mpesa transaction number	String Required
TransactionType	The transaction type being performed	String Optional
CommandID	CommandID for the specific business and transaction	String Required
CallBackChannel	The channel used to send the results of the transaction to the Bank. 1. Synchronous (default and only available if Mobile Money system is synchronus) 2. Async HTTP Post 3. Async SMS 4. Async eMail	String Required
CallBackDestination	Dependant on CallBackChannel Should be populated with relevant URL, MSISDN (format example 243821111111) or email address	String Required
InitiatorIdentifier	Bank's Identifier. M-Pesa team to provide.	String Required
SecurityCredential	Bank's Password. M-Pesa team to provide.	String Required

4.2 RESPONSE PARAMETERS

Name	Notes	Type	Sample Values
TransactionID	Unique Mpesa transaction number	String	Same as value sent in request
InsightTxID	The unique transaction ID from iPG	String	Same as value sent in request
TransactionDateTime	Date of the transaction. Format YYYYMMDD HH24MISS.	Date	20150716 150000
ResponseCode	Code indicating the outcome of the response.	String	0, -1, -2
ResponseDesc	Description of the response code provided. 0 – Received request successfully. -1 – Transaction will not be processed. Try again later. -2 – Duplicate Transaction	String	Received request successfully. Transaction will not be processed. Try again later. Duplicate Transaction

4.3 ASYNCHRONOUS RESULT DETAIL

4.3.1.1 RESULT (IPG TO BANK)

The details that will be sent back to the ResponseChannel from the Request packet once the asynchronous result comes into iPG from G2.

Name	Notes	Type	
TransactionID	The unique transaction ID on the mobile money platform	String	X-BS281
Amount	The transaction amount	String	120
TransactionTime	The date and time when the transaction was completed.	String	20170301T22:59:01
ThirdPartyReference	Third party transaction reference to iNSight	String	TestBank123456
InsightReference	The unique transaction ID from iPG	String	12sdxq3634dadsas242

ResultCode	<i>The result code of what happened with the transaction on the mobile money platform</i>	String	0
ResultDesc	<i>The result description of what happened with the transaction on the mobile money platform</i>	String	Process service request successfully.

4.4 SAMPLE ENVELOPES

4.4.1 Request

```
<Request>
  <Command>BankB2W</Command>
  <SessionID>cfb1b3b65d76438e83c6cd4fala17ab1</SessionID>
  <Shortcode>12345</Shortcode>
  <BankShortcode>100500</BankShortcode>
  <Currency>USD</Currency>
  <Amount>200</Amount>
  <DateTime>20150716 150000</DateTime>
  <ThirdPartyReference>TestBank123456</ThirdPartyReference>
  <TransactionID>123456</TransactionID>
  <CommandID>Initrans_XXXXX</CommandID>
  <TransactionType>VA</TransactionType>
  <CallBackChannel>2</CallBackChannel>
  <CallBackDestination>http://xxx.xxx.xxx:123/post</CallBackDestination>
</Request>
```

4.4.2 Response

6.4.2.1 Success

```
<Response>
  <ThirdPartyReference>TestBank123456</ThirdPartyReference>
  <InsightTxID>12sdxq3634dadsas242</InsightTxID>
  <TransactionDateTime>20150716150000</TransactionDateTime>
  <ResponseCode>0</ResponseCode>
  <ResponseDesc>Success</ResponseDesc>
</Response>
```

6.4.2.2 Duplicate Transaction

```
<Response>
  <ThirdPartyReference>TestBank123456</ThirdPartyReference>
  <InsightTxID>12sdxq3634dadsas242</InsightTxID>
  <TransactionDateTime>20150716150000</TransactionDateTime>
  <ResponseCode>-2</ResponseCode>
  <ResponseDesc>Duplicate Transaction</ResponseDesc>
</Response>
```

4.4.3 Result

6.4.3.1 HTTP Post (IPG to Bank)

To receive the result of an Asynchronous event call via an HTTP post, the `CallBackChannel` must be set to “2”. The `CallBackDestination` should also carry the URL that should be posted to from **iNSight** (it is important that **iNSight** can access this URL). The envelope posted will be a SOAP envelope. **iNSight** must be able to post without needing to authenticate or needing to perform any preliminary steps. The Bank must reply with the specified Response packet.

```
<Result>
  <TransactionID>X-BS281</TransactionID>
  <Amount>200</Amount>
  <TransactionTime>20170301T22:59:01</TransactionTime>
  <ThirdPartyReference>TestBank123456</ThirdPartyReference>
  <InsightReference>12sdxq3634dadsas242</InsightReference>
  <ResultCode>0</ResultCode>
  <ResultDesc>Process                service                request
  successfully.</ResultDesc> </Result>
```

6.3.2.1. HTTP Response (Bank to IPG)

```
<Response>
  <BankResponseCode>0</BankResponseCode>
  <BankResponseDesc>Received</BankResponseDesc>
</Response>
```

6.4.3.2 SMS

To receive the result of an Asynchronous event call via an SMS, the `CallBackChannel` must be set to “3”. The `CallBackDestination` should carry the MSISDN that the SMS must be sent to. The format of the SMS needs to be 2438xxxxxxx. Once the SMS is sent successfully, there will be no tracking of the receipt of the message by **iNSight**.

The format of the email will be as follows.

Body

```
TransactionID: [TransactionID]
Amount: [Amount]
TransactionTime: [TransactionTime]
ThirdPartyReference: [ThirdPartyReference]
InsightTxID: [InsightTxID]
ResultCode: [ResultCode]
ResultDesc: [ResultDesc]
```

6.4.3.3 Email

To receive the result of an Asynchronous event call via an email, the `CallBackChannel` must be set to "4". The `CallBackDestination` should carry the email address that the email must be sent to. Once the email is sent successfully, there will be no tracking of the receipt of the message by **iNSight**.

The format of the email will be as follows.

Subject [InsightReference] Transaction Completed

Body

```
TransactionID:[TransactionID]
Amount:[Amount]
TransactionTime:[TransactionTime]
ThirdPartyReference:[ThirdPartyReference]
InsightTxID:[InsightTxID]
ResultCode:[ResultCode]
ResultDesc:[ResultDesc]
```

5. RESPONSE ERROR CODE MAPPING

Code	Description	Comment
-6	Incorrect Parameter sent in B2B Request	One of the parameters being sent in the request is incorrect. Please check and try again.
-2	Duplicate Transaction received	Duplicate Transaction received
-3	B2B Pin Error Notification	B2B Pin Error Notification
-1	Invalid Login Credentials	Event user login details are invalid
-7	Technical Error on iPG	API failed due to technical reasons on IPG.
-9	Invalid Packet Structure/Content	Packet received doesn't match any configured events.
0	Success	Call/ process has been successful
-5	Subscriber Cancelled Pin Entry	Subscriber cancelled Pin Entry
-4	Subscriber Cancelled Pin Entry Retry	Subscriber cancelled Pin Retry
99	G2 Error code - G2 Error Description	G2 error response

6. RESULT ERROR CODE MAPPING

Code	Description	Comment
-1	Invalid Shortcode Provided	Provided shortcode does not match user
-2	No Response was Received from the Subscriber	No response was received from the Subscriber
-5	Subscriber cancelled the Pin entry	Subscriber cancelled Pin Entry
-3	Subscriber did not enter Pin	Subscriber did not enter Pin (Timeout)
-4	Subscriber cancelled the Pin entry retry	Subscriber cancelled Pin Retry
99	G2 Error code - G2 Error Description	G2 error response

If the payment to G2 initiated successfully the final notification Result code and description will depend on G2, we only pass the values they send to us.
